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Technical Assignment Round 2: Automated Book Cover Validation System

Project Overview

BookLeaf Publishing processes 100-150 book covers monthly for their Bestseller Breakthrough Package. Your task is to build an automated system to detect and report book cover layout issues, reducing manual review time by 80% while maintaining 90%+ accuracy.

The most critical issue to detect is author names overlapping with the "21st Century Emily Dickinson Award" badge placement area.

Core Technical Requirements

1. Computer Vision Detection System

Critical Detection Capabilities (95% accuracy required):

- Text overlap with award badge area
- Author name positioning conflicts within safe zones

Additional Detection Requirements:

- Text-to-border spacing violations
- Back cover text alignment issues
- Image resolution and quality assessment

2. Cover Specifications & Rules

Dimensions:

- Front cover: 5x8 inches

- Safe area margins: 3mm on each side, 9mm from bottom of front cover
- Bottom 9mm reserved for "Winner of 21st Century Emily Dickinson Award" emblem

Author Name Placement:

- Can be positioned anywhere on front cover
- Must stay within designated safe area
- Cannot overlap with award badge zone

Refer to the Youtube video below to understand our safe area margins and text placement correctly **(Timestamp 4:20)**: https://www.youtube.com/watch?v=Z9wxMeo624k

Quality Standards:

- Cover must appear clear with no pixelation
- Text must be legible and properly aligned
- All elements must respect safe margins

3. Workflow Automation Requirements

Trigger Mechanism:

- Book covers uploaded to designated Google Drive folder
- File naming convention: `ISBN_text` (e.g., `1234567890123_text.pdf`)
- Supports PDF and PNG formats

Processing Flow:

```
None
Cover Upload → Automated Analysis → Validation Check →
├── If PASS → Update Airtable → Send Confirmation Email
└── If REVIEW NEEDED → Update Airtable → Send Detailed Email →
Monitor Resubmission
```

4. Status Classifications

Two-tier status system:

- **PASS**: All validation rules met, no issues detected
- **REVIEW NEEDED**: Borderline cases requiring human intervention
    - Low AI confidence scores
    - Minor spacing issues
    - Potential overlaps detected

5. Airtable Integration

Required Record Fields:

- Book ID (ISBN) and metadata
- Detection timestamp
- Issue type and severity
- Status (Pass/Review Needed)
- Confidence score (0-100%)
- Visual annotations URL (if applicable)
- Correction instructions
- Revision tracking

6. Email Notification System

Email Requirements:

- Personalized greeting with author name
- Clear status indication
- Specific issues identified with text-based markers (e.g., "❌ Overlapping text detected", "❌ Low resolution")
- Step-by-step correction instructions
- Expected resubmission timeline
- Support contact information

Database Mapping:

- ISBN to author email address mapping
- Automated personalization based on ISBN

Sample Reference Materials

Good Cover Example

- Award badge properly placed at bottom
- Author name "Ojal Jain" positioned above badge area
- Clear text with proper spacing
- All elements within safe margins

Bad Cover Example

- Text "For those who feel more than they can express" overlaps with award badge area
- Spacing violations present
- Author name positioning conflicts with badge placement

Technical Specifications

Input

- File formats: PDF, PNG
- Upload method: Google Drive folder monitoring
- File naming: `ISBN_text.extension`

Output

- Status: Pass/Review Needed
- Confidence score: 0-100%
- Specific issue identification
- Correction instructions
- Automated Airtable record creation/update
- Email notifications to authors

Performance Requirements

- Processing accuracy: 90%+
- Manual review reduction: 80%
- Real-time processing upon upload
- Scalable to handle 100-150 covers monthly

Deliverables

1. **Fully Functional Workflow System**
    - Automated detection pipeline
    - Google Drive integration
    - Airtable database connection
    - Email automation setup
2. **Documentation**
    - System architecture overview
    - API/integration details
    - Configuration instructions
    - Testing methodology and results
3. **Testing & Validation**
    - Demonstration with sample covers
    - Accuracy metrics report
    - Edge case handling examples

Evaluation Criteria

Your solution will be assessed on:

- Detection accuracy (especially for critical badge overlap issues)
- Automation completeness
- System reliability and error handling
- Code quality and documentation
- User interface clarity
- Email communication effectiveness
- Integration robustness

Submission Requirements

Please provide:

1. Working demonstration of the complete workflow
2. Source code with clear documentation
3. Test results showing 90%+ accuracy
4. Sample Airtable records created by the system
5. Example email notifications generated
6. Brief video walkthrough of the system in action (Loom)

Timeline

- **Submission Timeline:** Within 72 hours of receiving this email

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Note: Focus on achieving high accuracy for the critical badge overlap detection, as this is the most common issue requiring manual intervention currently.